

Territorial dimensions of agro-environmental measures and LFA in rural development policy in the Czech Republic[☆]

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ABSTRACT

Outlined by the Common Agriculture Policy (CAP), the rural development policy enables the application of the territorial dimension. The goal of this paper is the analysis and determination of the appropriateness of the CAP development policy tools specifically within the Czech Republic. Furthermore, there is emphasis on the compliancy with the EU territorial cohesion objectives. The analyses concentrate on the agro-environmental measures (AEM) and less-favoured areas (LFA) support. In addition, payments during the 2004–2006 periods are evaluated. Relevant statistical indicators that include geographical, demographical, and economical are designated. Testing and verification of the indicators undertakes comparative examination based on municipal size-categories. The results of this process indicate that AEM and LFA schemes do not benefit the underdeveloped municipalities as intended by the EU territorial cohesion objectives.

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Introduction

The rural development policy faces a number of partial problems in its contextual definition and has an unclear position, particularly in connection with sectorial policies; for example transport, environmental and employment policy. The most important problem is still related to the prevailing perception of rural development with an emphasis on agriculture. This emphasis is specific to the EU (see Mahé and Ortalo-Magné, 1999; Lowe et al., 2002; Dwyer et al., 2007; Marsden and Sonnino, 2008). However, in the last two decades, the rural space has been significantly influenced by globalization trends, information technology improvements, reduction of transportation costs and the increase in the importance of non-agricultural activities in rural areas, etc. (OECD, 2006a, p. 12). These influences have had various impacts on rural areas. The definition of the rural areas in the EU used by the OECD is also problematic

and still under the “redefinition process”, while all these areas are different (Eurostat, 2010, p. 240, *A revised urban-rural typology*).

Until 2006, rural development in EU countries was influenced partially by the EU structural policy and by the growing second pillar of the EU Common Agricultural Policy (CAP). The first pillar of the gradually reformed CAP focuses on the direct support of the agricultural sector. The second pillar still contains conceptually unclear rural development. Within academic literature and at the policy-makers level in the EU, these are often considered to be two entirely separate pillars of the CAP, but the reality is different (Baldwin et al., 2008; Bureau, 2002). There is a certain overlap between the two pillars. It is the agricultural perception of rural development, i.e. particularly multifunctional agriculture support and the influence of its positive externalities on rural areas (Pelucha, 2009; Lapka et al., 2011).

Even under these circumstances, the rural development policy should maintain its complex, territorial and integral character (Léon, 2005; Wilkin et al., 2003). Moreover, the relation to territorial cohesion is defined as a specific horizontal objective of the EU's CAP rural development policy in paragraph 1 of the Council Regulation (EC) No. 1698/2005, about the support for rural development by the EAFRD, which says that “*rural development policy should also take into account the general objectives for economic and social cohesion policy set out in the Treaty and contribute to their achievement, while integrating other major policy priorities*”. Therefore the policy-making level needs to reflect and analyze the mentioned

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relation. The balanced territorial development in rural areas is also perceived by the European Commission (2010a, p. 7) as a key objective of the EU's CAP future 2014+.

The main goal of this paper is to analyze whether main rural development policy tools applied within the CAP in the Czech Republic are implemented in compliance with the territorial cohesion objectives of the EU. Particularly, we discuss the most important tools of the rural development policy in terms of their financial allocation and focus on land use. These are agro-environmental measures (AEM) and the support of less-favoured areas (LFA) that represents the above mentioned “agricultural” concept of rural development policy and strong emphasize on issues of land use planning. The paper is based on testing of AEM and LFA objectives that are straight and defined by the EU. The relation of these tools to territorial cohesion aspects (e.g. social stability, increasing new jobs) is seen as a positive externality. In the case of LFA, the payments support farmers in worse conditions from the point of view of natural conditions and therefore we can ask if there are any relations to broad socioeconomic issues of appropriate territory. Considering that the AEM focus on the support of relatively low intensity farming, the distribution of farmers and levels of subsidies per agricultural area are expected to be highest in the weaker rural regions.

The paper proceeds as follows. The second part represents a theoretical-methodological context of the definition of territorial cohesion and its relation to the rural development policy. In the third part, the role of the rural development policy in the CAP is assessed. The fourth part introduces the existing experiences in analyzing the relationship of the second pillar of the CAP with the aspects of territorial cohesion. The fifth section focuses on an analysis of the relationship of the AEM and LFA results (the largest tools of the EU's rural development policy) with the objectives of territorial cohesion. The final chapter summarizes key findings of the paper.

Definition of the “territorial cohesion”

Definition of the territorial cohesion is not entirely simple or unified. From the historical point of view, it is not a completely new term because it started to be used after the ratification of the Single European Act¹ in 1986, which brought new economic and social cohesion objectives to the EU policies. During this period, a more integrated approach to territorial development started to be applied while the restructuring of the agricultural sector and an increase in its competitive advantage started to be considered part of the complex rural development (Marsden and Bristow, 2002). The necessity of discussing territorial cohesion at the level of EU authorities was intensified by the last EU enlargement in 2004 and 2007, accompanied by the increase of regional disparities, and the increase in the importance of the single market. Leonardi (2006) considered the relevant tools of regional policies focused on territorial cohesion as “shock absorbers”.

According to Shucksmith et al. (2005), the term of territorial cohesion is defined much broader than economic and social cohesion. He believes that *the objective of territorial cohesion is defined as helping to achieve a more balanced development by reducing existing disparities, preventing territorial imbalances and by promoting greater coherence between both sectoral policies that have spatial impacts and regional policy* (Shucksmith et al., 2005, p. 11). This implies that territorial cohesion is a complex issue of which purpose is to

horizontally and vertically connect the stakeholders and natural processes of regional development. The term territorial cohesion represents a dynamic view of the observed phenomena; i.e. certain effort to reach a balance between territorial development and reduction of the most important regional disparities in the socio-economic development.

In 2008, the European Commission ratified the Green Paper on Territorial Cohesion (European Commission, 2008) in which territorial cohesion is defined as a prism of territorial diversity in the EU that should become an advantage in providing for sustainable development to the entire EU. Territorial cohesion is understood as a process through which “harmonious” development should be achieved. However, we believe that the definition mentioned in the Green Paper on Territorial Cohesion is terminologically too broad. The fifth cohesion report (European Commission, 2010b, p. 7) explains only the possible focus of the territorial cohesion as a third dimension of the structural policy, e.g. the role of cities, functional geographies, areas facing specific geographical or demographic problems and macro-regional strategies.

According to the definition of the European Commission, we encounter the problem of measurability of territorial cohesion, especially the identification of certain factors that would prove the positive and negative aspects of this process. Positive aspects would be based on the stability of territorial development. Negative aspects of the territorial cohesion process can be found in the efforts to fully cover all components of socioeconomic development, which is conceptually hard to seize and measure. The mentioned definition's issues suggest that this is a rather normative concept that is not fully matched by its theoretical model.

It is necessary to stress that the assessment of the rural development policy and its relation to territorial cohesion is a problem. In defining the territorial cohesion and its objectives we used the study of Shucksmith et al. (2005), in which the extent of dependence between the CAP rural development policy aid and selected socio-economic indicators were tested at the NUTS 3 regional level in the EU-15 countries.

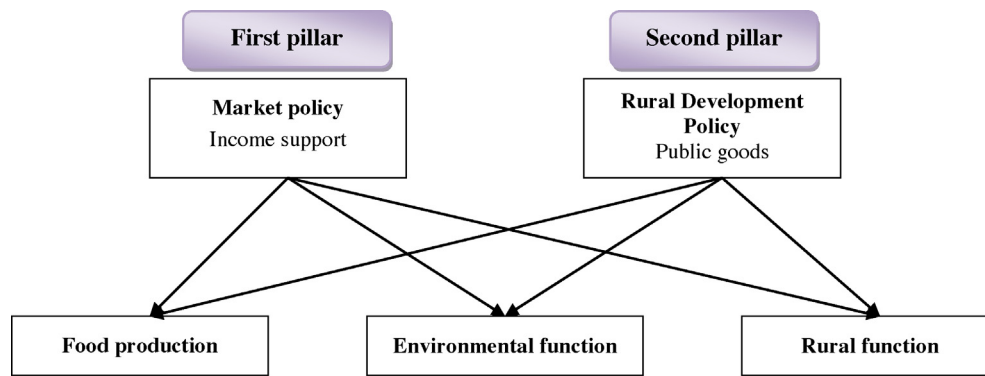
Declared and factual role of the rural development policy in the CAP

At the launch of the EU's CAP, the support of non-agricultural activities was low (see e.g. Lowe et al., 2002; Dwyer et al., 2007; Baldwin et al., 2008). McSharry's reforms² and Agenda 2000³ were the first strategic steps that officially declared rural development as a key component of the CAP. The result of the Agenda 2000 was a division of the sector-based CAP into two pillars; (1) aid for production agriculture on one hand and (2) the newly created rural development policy on the other. Both pillars should be in a specific balance. The concept of this balance is demonstrated in Fig. 1 which represents the European Commission's idea for the ideal balance of the CAP. The first pillar is aimed at agriculture production support (food security and availability). McAreevey (2009, p. 22) stated that this pillar is quite complicated in terms of current “*evidence that current subsidies-based policies are ineffective in addressing socio-economic challenges facing rural communities*”. On the other hand is the rural development policy, which is considered as a territorially based pillar with a specific target to eliminate negative impacts of intensive agriculture farming and to diversify socio-economic structure of rural areas.

¹ The Single European Act significantly modified and amended the Treaties of Rome of 1957. It was signed in February 1986 by twelve member states of the European Community and came into force on 1st July 1987. For more details on this document see EURLEX (2009).

² The CAP was first officially reformed in 1992 and the reform was titled after the Commissioner for Agriculture Ray McSharry. At that time, it was his approach and implementation of several reform points that helped during the negotiations on the form of the CAP within the WTO.

³ Agenda 2000 was approved by the European Council in 1999.



Source: European Communities (2004)

Fig. 1. Declared balance between the pillars of the EU's CAP.

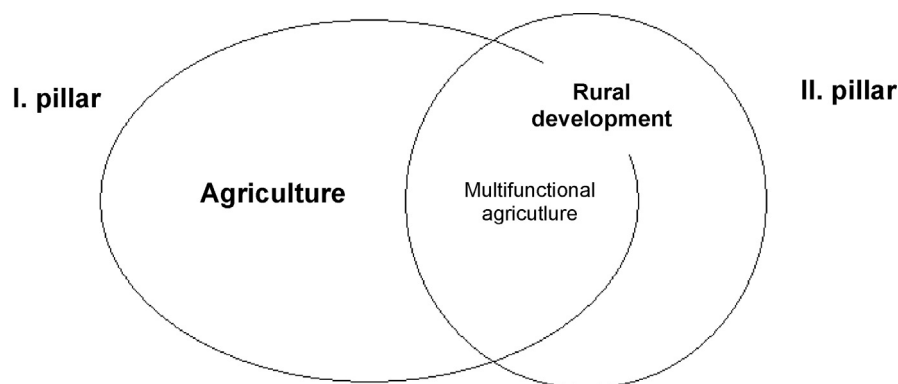
Source: European Communities (2004).

The involvement of rural development policy into the CAP was caused by two factors. Firstly, the CAP started to be under significant pressure during the 1990s, when the critique of this policy was rising due to its inefficiency, inadequate redistribution impacts and increasing financial expenditures (Balmann, 2006; Marshall, 2001). Secondly, the CAP became the target of criticism during negotiations with the World Trade Organization (WTO) which did not favour the protectionism under this policy and its negative impacts on international trade in agricultural commodities (Bureau and Mahé, 2008; Swinbank, 1999). In this regard, Tangermann (2006) considered the involvement of the rural development policy into the CAP as a “jink” during discussions at the WTO level on further liberalization of the agricultural market. In this sense, the second pillar should help to maintain a more significant allocation of agricultural sector funds. However, the rural development is a more complex and territorially based issue (Barca, 2006; Caballero et al., 2005). Moreover, Ward and Brown (2009) emphasized that *sectoral policies, such as those focusing on supporting agriculture, are increasingly inappropriate and ineffective in stimulating rural development* (Ward and Brown, 2009, p. 1238).

Structural aspects of the rural development policy should shift the CAP in the area of its second pillar towards enhancement of territorial cohesion and land use planning issues. But there are several factual problems. According to Bureau (2002, p. 84), the focus of most financial sources of the CAP towards agricultural producers is a limiting factor for enhancement of socio-economic objectives of territorial cohesion. Janvry

et al. (1998) postulates other limiting factors, e.g. the structural characteristics of rural areas themselves that cause social and ecological problems (distance, dispersion, activities based on natural resources, imperfect legislation on proprietary rights, persisting market failures, intersectoral allocation of resources of urbanistic policies and imperfect regional co-operation, etc.). Therefore, there is the question, how factual are the relationships between such an EU rural development policy and territorial cohesion.

At the given moment, we cannot talk about two completely separate pillars of the CAP that supplement each other as it is presented in Fig. 1. There is a large overlap in understanding the “rural development” in its agricultural concept (see Fig. 2). In literature (e.g. Bureau, 2002; Wilson, 2007; Marsden and Sonnino, 2008; Pelucha et al., 2009) and strategic documents (OECD, 2001, 2006a; European Commission, 2007) this overlap is referred to as so-called multifunctional agriculture that should influence rural areas through its “positive externalities”. According to OECD (2006b, p. 15), the degree of overlap means that coherence between agricultural and rural policies is essential because of the role of farming which is the dominant user and manager of land. Furthermore, Woods (2011, p. 80) stated that *the concept of multifunctionality has been adopted more widely within rural geography and rural sociology, it has come to mean more than simply a differentiated agricultural economy. In particular, it has come to refer to the multiple outcomes of agriculture, which include not only the production of food and other resources, but also social and environmental benefits.*



Source: Own work

Fig. 2. Current state of the CAP with two pillars and the concept of rural development with an emphasis on its agricultural dimension in view of multifunctionality. Source: Own work.

Rural development, currently in the CAP, is still perceived as agricultural-focused, and not as a complex policy. The big question remains the same: whether aid for multifunctional agriculture can enhance territorial cohesion in EU regions. Johnson (2001) clearly states that *agricultural policy may not be considered rural development policy* (Johnson, 2001, p. 33). Based on his analysis, he critically discussed the situation of agricultural support in the USA, which is also at a high level, just like in the EU. Nevertheless, its positive influence on improvement of quality of rural life is indirect, weak and ambiguous.

Existing analyses of the EU's CAP second pillar relation to the aspects of territorial cohesion

In existing analyses, there has not been proven a close relation of the rural development policy in the CAP second pillar to the aspects of territorial cohesion (e.g. Shucksmith et al., 2005; Viktorová and Pělucha, 2005; Hrabankova et al., 2006; Stolbova, 2007; Montresor, 2008). Reasons for these results partially stem from the diverse factual set-up of rural development policies at the level of EU member states, which is a consequence of certain institutional conservatism (Dwyer et al., 2007).

So far, the most important and most extensive analysis of the relation of the CAP second pillar to the aspects of territorial cohesion was carried out by Shucksmith et al. (2005), who analyzed this issue by simple Pearson linear correlations during the years 2002–2004. By the way, in his current works, e.g. EDORA research project (Dax et al., 2010), Shucksmith stressed the description of process using such terms as “grand narratives” than doing a statistical analysis. On the other hand, there is a strong demand of the European Commission for the quantitative approaches to policy-making and evaluation methods (e.g. counterfactual analysis using the regression and correlation analyses).

Shucksmith et al. (2005) based his analysis on the hypothesis that the second pillar is a specific instrument of the CAP with a territorial dimension and should be focused on an integrated approach to rural development. In this sense, the activities of the second pillar should be supported in compliance with the territorial cohesion objectives. Because of the unavailability of data, in the assessment they focused on the most important tools of the CAP second pillar for the period 2000–2006, i.e. LFA scheme and AEM.

Key comparisons were reached through statistical correlation analysis of data from the selected year 2003 in all NUTS 3 regions across EU-15 countries. With respect to the size of these regions, payments in absolute numbers must have been converted to utilized agricultural area (UAA) and work units in agriculture (AWU). A correlation analysis of the above payments utilized socio-economic ratio GDP per capita, unemployment rate, relative change in population (1989–99).

The relevant results of the correlation analysis of the total payments under the second pillar of the CAP and the territorial cohesion objectives were surprising in total relations and rather conflicting in a relationship to territorial cohesion. The support of the second pillar of the CAP did not appear to be consistent with the territorial cohesion objectives at the regional level (NUTS 3); not even in the most detailed analyses of LFA and AEM payments. Shucksmith et al. (2005, p. 70) expressed that *their correlation analysis found no statistically significant relationships between levels of LFA and AEM support and indicators of economic cohesion, although the signs of the coefficients were as expected. Therefore, LFA and AEM support are only weakly related to the indicators of social and economic cohesion.*

The discrepancy between the key measures of the rural development policy and the territorial cohesion objectives is caused particularly by different national priorities in the CAP second

pillar. Dwyer et al. (2002, 2007) added other reasons to this problem, i.e. the uneven allocation of rural development financial sources, and difficulties in co-financing of rural development expenditure in poorer countries. Shucksmith et al. (2005, p. 67, 70, 71) summarizes that the LFA scheme and AEM are more prioritized in the richer EU states, while the rest of rural development measures are used in poorer states and regions. By this fact, Shucksmith et al. explained why a stronger relationship with the cohesion indicators was not detected. Partially, this finding was also confirmed by Ramniceanu and Ackrill (2007, p. 429) who summarized that EU member states allocate their public funds to multifunctional agricultural measures if they are richer and with a smaller and efficient agriculture sector.

According to above mentioned problems of the Shucksmith's methodology, it is possible to conclude the following modification of the described approach:

- The analysis approach of Shucksmith et al. (2005) should be applied on the level of individual member states rather than across the entire territory of the EU. The relative importance of rural development measures varies widely between the EU member states in terms of their emphasis on different national priorities and national budget constraints (Dwyer et al., 2002, 2007). Additionally, there are significant differences between old and new EU member states (Gorton et al., 2009).
- Considering the diverse character of rural areas defined at the NUTS 3 level, this approach should be applied only at the local or microregional level and separately by each EU member state. NUTS 3 are still large inhomogeneous regions that include cities and larger urban areas, therefore the best level for these analyses are municipalities (LAU II⁴).
- It is not appropriate to use data for only one year in statistical analyses. To eliminate possible fluctuations in the given year, statistical analyses should include a longer time period, at least three consecutive years which usually represents half of a programming period.
- It is desirable to extend the comparative analyses by other relevant socio-economic indicators. In similar research projects, there is a general problem with the availability of statistical data at the regional level. On the other hand, national statistics usually provide a relatively wide range of socio-economic statistical indicators⁵.
- Statistical analyses should contain data on actual payments – i.e. expenditures of the given measure actually spent – not predicted budgets that can change over time. It is better to prove the relationship of the rural development policy with cohesion objectives by real expenditures in appropriate measure.

We applied all the above partial recommendations on the example of the Czech Republic – see the following part. The added value of this paper is the usage of relevant data at the LAU II level. This local level is more relevant for the assessment of rural development policy compliance with territorial cohesion objectives. Furthermore, we used expenditures under the given measures and not only predicted budgets as it was presented in Shucksmith's approach.

⁴ LAU (Local Administrative Units) – represents the lowest level of territorial-administrative division.

⁵ In the case of the Czech Republic, e.g. even data on education (PHC – Population and Housing Census, 2001) and economic activity of population (if divided to the primary sector level) are available at the lowest level LAU II.

Analysis of the relationship between the CAP second pillar and the territorial cohesion objectives on the example of the Horizontal Rural Development Plan – Czech Republic

AEM and LFA scheme are the most important measures of the EU rural development policy. Their objectives should be closely related to territorial cohesion objectives. LFA payments are supporting farmers in worse conditions from the point of view of natural conditions. The AEM focus on the support of relatively low intensity farming and the distribution of farmers and levels of subsidies per agricultural area are expected to be highest in the weaker rural regions. According to these objectives, OECD (2006b, p. 19) stated that “agriculture is more than just the production of food and fibre – it is also associated with the provision of countryside amenities, and with positive and negative environmental outcomes, in particular through its dependence on land and water resources.”

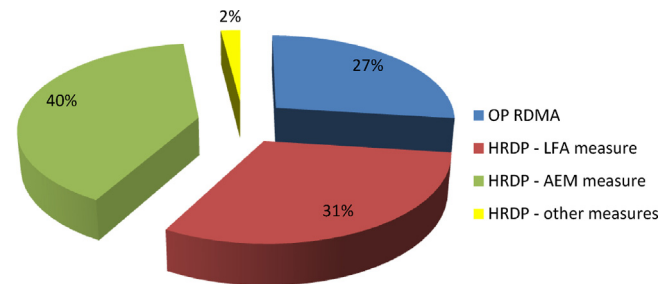
These measures represent the agricultural view of the rural development and started to be implemented in the Czech Republic within the Horizontal Rural Development Plan (HRDP) during the programming period 2004–2006. The short programming period 2004–2006 symbolized the entry step for implementation of the EU rural development policy in the Czech Republic. During this period, the CR started to implement the CAP second pillar as a regular EU member state. The HRDP was one of the key financial sources of rural development subsidies. Concurrently with the HRDP in 2004–2006, there was also implemented the OP RDMA – the Operational Programme Rural Development and Multifunctional Agriculture. Both programmes together represented sources of EU rural development funding (the second pillar of the CAP).

While the HRDP were aimed on the support of environmental nature protection in rural areas with partial socio-economic impact, the OP RDMA was focused more on the support of agricultural investment activities (e.g. restructuring and modernization of agriculture, forest management, diversification of agricultural activities and agriculture-related activities, LEADER+).

The concept and basic conditions of HRDP's measures (Early Retirement from Farming, Forestry, Setting up Producer Groups, AEM, LFA) were formulated in such a way that there should be a certain relationship between the aspects of territorial cohesion and the LFA and AEO measures. In terms of financial resources, the HRDP was clearly the largest and most important program within the second pillar of the CAP during the period 2004–2006 in the CR compared to the OP RDMA. Moreover, there was a clear tendency in the Czech Republic to prioritize the AEM support (EUR 365.5 m) and LFA support (EUR 283.7 m), which are of higher significance than the entire OP RDMA (EUR 250.6 m). This fact can be clearly seen in Fig. 3. Approximately 73% of all expenditures related to the CAP second pillar were spent through the HRDP. Therefore, the analysis of the implementation of the LFA and AEM measures and their relationship to the aspects of territorial cohesion is of great significance, because the share of these measures was important.

LFA payments in the context of territorial cohesion

The LFA measure is closely related to the support of “multi-functionality” of agriculture; especially in the disadvantaged areas. In the CR, LFA payments represent a regionally bound social payment with environmental restrictions (related only to permanent grass areas and authorized concentration of animals, etc.). Generally, the relationship of the LFA support to territorial cohesion is seen by Shucksmith et al. (2005, p. 101) in maintaining the farm management in less-favoured areas based on environmental



Source: Own processing of data – Ministry of Agriculture (2004a), Ministry of Agriculture (2004b)

Fig. 3. The structure of resources allocated from the second pillar of the CAP in the Czech Republic in 2004–2006.

Source: Own processing of data – Ministry of Agriculture (2004a) and Ministry of Agriculture (2004b).

principles and provision of other functions beyond food production.

In accordance with our suggestions on the modified methodology, we have carried out statistical comparisons (Pearson's correlation coefficient) of the key socio-economic indicators and the data on total LFA payments (within the HRDP). We analyzed selected statistical indicators and LFA payments on the level of municipalities (LAU II) in the CR. This approach increased the significance and importance of our results, because an assessment of tendencies in the issue of rural development has the highest relevant value at the level of municipalities than NUTS 3 level. The chosen method of correlation analysis surely enables us to evaluate the dependence of individual variables. We do not focus on seeking causality (for that would be better to use more sophisticated econometric methods), but we tried to clarify statistical relations between subsidies and selected socioeconomic indicators.

We analyzed all LFA payments made during the period 2004–2006 (data from SAIF, 2008) in comparison to the wide range of relevant socio-economic indicators. We used the following groups of indicators for the analysis:

1. Policy indicators

- LFA subsidies at the level of municipalities during the period 2004–2006: absolute value of all payments in the given municipality (SAIF, 2008);
- LFA subsidies at the level of municipalities during the period 2004–2006 converted to agricultural area (SAIF, 2008);
- Note: Both indicators are based on real expenditures and not on budget figures as it was used by Shucksmith et al. (2005).

2. Geographical indicators

- Area of the municipality: Czech Statistical Office (CSO), data from 2006. This indicator is crucial for verification of the precondition that larger municipalities receive higher payments;
- Agricultural area: CSO, data for 2006 at the level of municipalities. This indicator presents the intensity of agricultural land-use in appropriate municipality.
- Altitude: CSO, data from 2006 at the level of municipalities. Altitude presents the delimitation of LFA and also AEM payments, and therefore it is useful to check its relation to the broader context of territorial cohesion objectives;

3. Demographical indicators

- Relative migration balance: CSO, the migration balance indicator for the entire period 2004–2006 at the level of municipalities. The relative migration indicator indicates the ratio of increase/decrease in population of such municipality caused by migration;

- Population by education⁶: CSO, PHC 2001 at the level of municipalities (primary, secondary and university level education). We used this indicator for analyzing the level of qualitative development of the municipality;
- Population: CSO, data for 2006 at the level of municipalities. Rural municipalities are usually small in terms of their population. The number of inhabitants is important in terms of discovering if the less populated municipalities receive higher or lower level of payments;

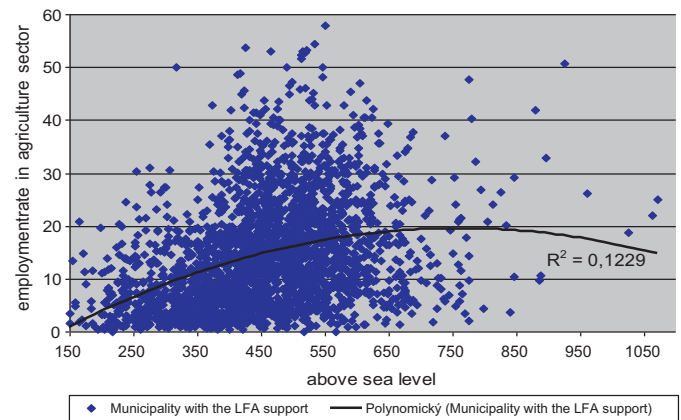
4. Economic indicators

- Number of job opportunities per 100 persons (NJO): based on CSO data that were recalculated according to the Hampl's NJO equation (HAMPL, 2007).⁷ We did not include the indicator related to GDP per capita into the above set of socio-economic indicators. In the Czech Republic, the most detailed GDP indicator is reported at the regional level, i.e. NUTS 2 (cohesion regions) and NUTS 3. For this reason we have selected other alternative indicators that denote economic opportunities.
- EAP: a ratio of the economically active population to the total population of the municipality – CSO, data for 2006. The indicator is crucial due to a possibility to analyze the relation of subsidies and general EAP including non-agricultural activities as well;
- EAP prim: a ratio of the total economically active population in the primary sector (i.e. agriculture, forestry and fishery) to the population of the municipality – CSO, data for 2006. This indicator presents the level of agricultural activities in the municipality;
- Unemployment rate: CSO, data for 2006 at the level of municipalities. We consider the unemployment rate as an indicator of economic success and municipal attractiveness;

Table 1 presents correlation coefficients between the LFA payments (as well as adjusted LFA payments) at the level of municipalities and the above set of socio-economic indicators. LFA subsidies funded 2646 municipalities, which represent approximately 42% of all municipalities in the Czech Republic.

Positively significant correlations are in bold, negatively significant correlations are in light blue. Table 1 shows different results in terms of absolute and adjusted LFA payments. While the absolute LFA payments are strongly correlated to the area of the municipality, population and agricultural area, the adjusted LFA payments per agricultural area only show weak correlation. This result leads to the logical conclusion that larger and more populated municipalities received higher payments. This finding confirmed the fact that the main part of the LFA scheme was distributed into larger municipalities with good conditions for agricultural and forestry activities. This particular important result is a confirmation of a long-term Czech Republic's characteristic which was identified by Doucha and Vanek (OECD, 2006b, p. 245): "LFA payments are largely absorbed by extremely large farms (often in the hands of non-agricultural/foreign capital), with a very low level of labour inputs, which maintain the countryside but do not increase its diversity. These farms are real winners of EU accession, but their beneficial effects on well-being in the rural community are small."

Correlation analysis found no statistically significant relationship between levels of LFA payments and the unemployment rate



Source: Own processing of data from SAIF (2008), CSO (2008)

Fig. 4. The relationship between the unemployment rate in the primary sector and altitude of municipalities receiving LFA subsidies in the Czech Republic (2004–2006).

Source: Own processing of data from SAIF (2008) and CSO (2008).

and the relative migration balance (either for absolute, or for data per agricultural area). As per Table 1, it is not possible to prove a significant relationship between LFA payments and the level of education of the population in the given municipalities. Negative correlation coefficients which are highlighted in the table (population with primary education) could serve as an interesting result for further research, only if these coefficients would not be independent. In this regard, the results of the Czech Republic are similar to the results of Shucksmith et al. (2005, p. 70) where this relationship was also not confirmed across the entire EU at the level of NUTS 3 regions. Basically, we reached a result that LFA payments were not awarded to underdeveloped or less economically developed municipalities.

Comparison of municipalities with and without LFA payments

Despite the above results we compiled Table 2 which contains average values of selected socio-economic indicators that are further divided according to various municipality size categories. In this table, we included all municipalities in the Czech Republic except for the capital Prague, which would otherwise highly distort the average values. We set the population over 3000 as a limit, because this number is considered a threshold value for determining whether such municipality is a town or not (according to the Act No. 128/2000 Coll., on Municipalities, § 3, section 1).

This section of the analysis compares the situation of smaller and larger municipalities that benefit from LFA payments. The results of Table 2 confirm the basic conditions stemming from the definition of this measure. It concludes that smaller municipalities with LFA subsidies had a higher altitude and lower subsidies per agricultural area. According to the individual categories, the unemployment rate was not higher in municipalities receiving LFA subsidies than in municipalities not receiving these funds. The rates of an economically active population, as well as the primary sector employment rate, were on average, higher in municipalities with LFA subsidies than in municipalities without LFA subsidies. In terms of the total migration rate, municipalities with LFA subsidies are more stable than municipalities without the latter. Therefore, the average values of the selected indicators confirm the results of the statistical correlation analysis. The correlation coefficient (0.331) between altitude and employment in the primary sector confirms the conditions set for LFA applicants. This relationship indicated in Fig. 4 suggests that municipalities with a high altitude are typical with a strong agricultural sector in terms of providing job opportunities. In the context of cohesion issues, we conclude that LFA payments

⁶ Even though the educational structure has surely changed since 2001 with regard to a higher availability of third level education in particular, other similar data at the level of municipalities are not available. Last PHC was carried out in 2011 and appropriate data at the municipal level were not available at the time of writing this article. In this sense, we assume that the composition of division of this index at the level of municipalities is stable.

⁷ i.e. Number of Job Opportunities (NJO)=Economically active Population (EAP) – number of unemployed persons – persons commuting for work out of the region + persons coming to work in the given area (HAMPL, 2007).

Table 1

Pearson correlation coefficient between LFA subsidies per agricultural area and selected socio-economic indicators (LAU II level).

	LFA subsidies	Area of the municip.	NJO per 100 persons	Relative migration balance	Altitude	Popul. elementary education	Popul. secondary education	Popul. university education	EAP	EAP prim	Unempl. rate	Population	Agricultur. area	LFA subsidies per a.a.
LFA subsidies	1	0,571(**)	0,154(**)	-0,070(**)	0,096(**)	-0,101(**)	0,058(**)	0,079(**)	0,139(**)	-0,116(**)	0,067(**)	0,533(**)	0,571(**)	xxx
Area of the municip.	0,571(**)	1	0,241(**)	-0,094(**)	0,017	-0,170(**)	0,121(**)	0,150(**)	0,189(**)	-0,167(**)	0,113(**)	0,608(**)	0,857(**)	0,042(*)
NJO per 100 persons	0,154(**)	0,241(**)	1	0,013	-0,042(*)	-0,177(**)	0,211(**)	0,156(**)	0,149(**)	-0,139(**)	-0,040(*)	0,094(**)	0,213(**)	0,266(**)
Rel. migr. balance	-0,070(**)	-0,094(**)	0,013	1	-0,078(**)	-0,422(**)	-0,143(**)	-0,107(**)	-0,530(**)	-0,114(**)	-0,033	-0,025	-0,098(**)	-0,021
Altitude	0,096(**)	0,017	-0,042(*)	-0,078(**)	1	0,051(**)	-0,197(**)	-0,115(**)	-0,033	0,331(**)	0,031	-0,114(**)	-0,079(**)	0,115(**)
Popul.element. educ.	-0,101(**)	-0,170(**)	-0,177(**)	-0,422(**)	0,051(**)	1	-0,152(**)	-0,148(**)	0,422(**)	0,169(**)	-0,074(**)	-0,097(**)	-0,134(**)	-0,073(**)
Popul. second. educ.	0,058(**)	0,121(**)	0,211(**)	-0,143(**)	-0,197(**)	-0,152(**)	1	0,425(**)	0,339(**)	-0,215(**)	-0,338(**)	0,140(**)	0,130(**)	0,018
Popul. univ. educ.	0,079(**)	0,150(**)	0,156(**)	-0,107(**)	-0,115(**)	-0,148(**)	0,425(**)	1	0,209(**)	-0,258(**)	-0,166(**)	0,133(**)	0,144(**)	0,015
EAP	0,139(**)	0,189(**)	0,149(**)	-0,530(**)	-0,033	0,422(**)	0,339(**)	0,209(**)	1	-0,052(**)	-0,086(**)	0,072(**)	0,183(**)	0,055(**)
EAP prim	-0,116(**)	-0,167(**)	-0,139(**)	-0,114(**)	0,331(**)	0,169(**)	-0,215(**)	-0,258(**)	-0,052(**)	1	-0,013	-0,107(**)	-0,153(**)	-0,049(*)
Unempl. rate	0,067(**)	0,113(**)	-0,040(*)	-0,033	0,031	-0,074(**)	-0,338(**)	-0,166(**)	-0,086(**)	-0,013	1	-0,013	0,049(*)	0,052(**)
Population	0,533(**)	0,608(**)	0,094(**)	-0,025	-0,114(**)	-0,097(**)	0,140(**)	0,133(**)	0,072(**)	-0,107(**)	-0,013	1	0,572(**)	0,021
Agricultural area	0,571(**)	0,857(**)	0,213(**)	-0,098(**)	-0,079(**)	-0,134(**)	0,130(**)	0,144(**)	0,183(**)	-0,153(**)	0,049(*)	0,572(**)	1	-0,006
LFA subsidies per agri.area	xxx	0,042(*)	0,266(**)	-0,021	0,115(**)	-0,073(**)	0,018	0,015	0,055(**)	-0,049(*)	0,052(**)	0,021	-0,006	1

* Statistical significance at the 95% confidence level

** Statistical significance at the 99% confidence level

Source: own processing of data from SAIF (2008), CSO (2008)

Source: Own processing of data from SAIF (2008) and CSO (2008).

*Statistical significance at the 95% confidence level.

**Statistical significance at the 99% confidence level.

Table 2
Comparison of selected indicators for municipalities with LFA subsidies and municipalities without LFA subsidies according to selected size categories.

Municipality size category (residents)	Municipalities with LFA subsidies from HRDP						Municipalities without LFA subsidies from HRDP							
	Relative migration balance (%)	EAP (%)	EAP primary sector (%)	Unempl. rate (%)	Average subsidies (CZK)	Average area (ha)	Subsidies/ Agr. area	Average altitude	Relative migration balance (%)	EAP (%)	EAP primary sector (%)	Unempl. rate (%)	Average area	Average altitude
Less than 200	1.28	4739	2281	8.94	578,174.35	648.48	2048.54	502.52	2.14	4668	1808	9.94	557.67	398.57
200–500	1.95	4764	1621	8.95	1,248,176.14	1170.11	2244.02	473.05	3.08	4706	1330	9.43	731.14	330.23
500–1000	1.94	4784	1245	9.13	2,098,219.49	1761.90	2448.39	448.40	3.42	4742	905	8.96	992.82	293.16
1000–3000	1.99	4861	872	8.50	3,521,589.39	2557.80	3088.52	422.98	3.04	4783	719	9.19	1476.84	266.07
More than 3000	0.15	5150	361	8.48	5,432,577.34	4273.94	3081.90	380.36	1.23	5023	336	9.41	2509.10	258.73

Source: Own processing of data from SAIF (2008) and CSO (2008).

are directed at richer or less problematic areas. There is a partial exception that the level of the rates of an economically active population and employment in the primary sector are on average in all municipalities with LFA payments higher than in municipalities without these payments. [Appendix No. 1](#) displays a cartogram of LFA payments per agricultural area in which is presented the situation of supported municipalities (LAU II). In geographical terms, the area covers mainly highland regions with respectable conditions for pastoral farming.

The Agro-environmental measures in the context of territorial cohesion

According to [Shucksmith et al. \(2005, p. 155\)](#), the relationship of the AEM to territorial cohesion should be seen in the fact that the AEM have multiple objectives. As eligibility to participate tends to be contingent on relatively low-intensity farming, and therefore the distribution of participants and levels of payment per AWU and per UAA are expected to be highest in the weaker rural regions. Therefore, AEM are expected to contribute to the achievement of the economic and social cohesion goals. AEM is used by farmers located in areas with less favourable conditions for agriculture where the protection of landscape is at a higher level of sensitivity than in other traditional agricultural regions. [Brouwer and Heide \(2009, p. 340\)](#) added that the poor economic viability of these marginal farmlands implies that targeted financial support – that is public payment – is required if their sensitive biodiversity management is to be attained. However, AEM have often been roundly criticized for not achieving their objectives. In the area of nature and landscape, this can be quite complicated because public policy objectives are often not agreed through quantitative goals.

We analyzed the AEM payments using the same method as in the case of the LFA measure. The methodological aspects were mentioned in the previous part. [Tables 3 and 4](#) present the correlation between absolute AEM payments as well as AEM payments per agricultural area and selected socio-economic indicators. In total, we included 3509 municipalities where applicants for AEM subsidies were located, which represents approximately 56% of all municipalities in the Czech Republic. Again, positively significant correlations are in bold, while negatively significant correlations are in light blue.

As with the case of the LFA payments, we calculate various correlation coefficients in the case of absolute AEM payments and AEM payments per agricultural area. [Table 3](#) reflects this relationship. With regard to the set-up of the AEM, there is a strong correlation between the absolute payments and the area of the municipality, population, and the agricultural area. These findings confirm logical issues, in that larger municipalities receive a higher amount of allocated financial resources. The amounts of AEM subsidies per agricultural area then indicate a slightly positive correlation (0.157). The number of job opportunities however, is low and are close to the statistical independency of the two indicators. Comparable to [Shucksmith et al. \(2005, p. 70\)](#), the case of the Czech Republic does not confirm a strong relationship to the aid allocated to regions with worse socio-economic development conditions. Possible reasons for such results are particularly suggested by [Doucha and Vanek \(OECD, 2006b, p. 245\)](#) that AEO measures are mainly preserving a “status quo” in the relationship between agriculture and environment and the changes which were necessary to make to land-use are not being sufficiently stimulated.

In [Table 3](#) (just like in the case of LFA payments) a negative relationship between the level of education and the AEM subsidies was not confirmed. The only significant difference between the results based on the data from the Czech Republic and the situation in the EU regions ([Shucksmith et al., 2005, p. 71](#)) is

Table 3
Pearson correlation coefficient between AEM subsidies per agricultural area and selected socio-economic indicators (LAU II level).

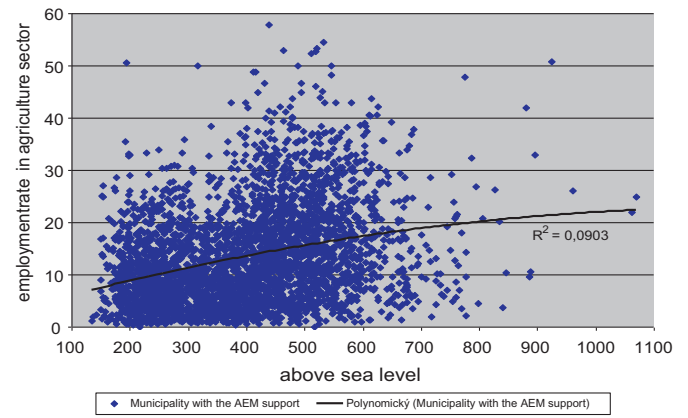
Municipality size category (residents)	MUNICIPALITIES with LFA subsidies from HRDP								MUNICIPALITIES without LFA subsidies from HRDP					
	Relative migration balance (%)	EAP (%)	EAP primary sector (%)	Unempl. rate (%)	Average subsidies (CZK)	Average area (ha)	Subsidies / Agr. area	Average altitude	Relative migration balance (%)	EAP (%)	EAP primary sector (%)	Unempl. rate (%)	Average area	Average altitude
Less than 200	1,28	47,39	22,81	8,94	578 174,35	648,48	2 048,54	502,52	2,14	46,68	18,08	9,94	557,67	398,57
200 - 500	1,95	47,64	16,21	8,95	1 248 176,14	1 170,11	2 244,02	473,05	3,08	47,06	13,30	9,43	731,14	330,23
500 - 1000	1,94	47,84	12,45	9,13	2 098 219,49	1 761,90	2 448,39	448,40	3,42	47,42	9,05	8,96	992,82	293,16
1000 - 3000	1,99	48,61	8,72	8,50	3 521 589,39	2 557,80	3 088,52	422,98	3,04	47,83	7,19	9,19	1 476,84	266,07
More than 3000	0,15	51,50	3,61	8,48	5 432 577,34	4 273,94	3 081,90	380,36	1,23	50,23	3,36	9,41	2 509,10	258,73

Source: own processing of data from SAIF (2008), CSO (2008)

Source: Own processing of data from SAIF (2008) and CSO (2008).

*Statistical significance at the 95% confidence level.

**Statistical significance at the 99% confidence level.



Source: own processing of data from SAIF (2008), CSO (2008b)

Fig. 5. The relationship between employment in the primary sector and altitude in municipalities that receive AEM subsidies in the Czech Republic (period 2004–2006). Source: Own processing of data from SAIF (2008) and CSO (2008).

the relationship between subsidies per agricultural area and the unemployment rate. The data on EU regions proved this relationship to be negative and thereby indirectly demonstrated surprising information – AEM payments are mostly distributed into less problematic EU regions. In the case of the Czech Republic, the data on AEM payments and the unemployment level seem to be independent. The correlation coefficient related to the number of job opportunities in connection with AEM subsidies per agricultural area is 0.157 which can be considered as a low correlation coefficient.

Comparison of municipalities with and without AEM subsidies

Even though more significant relationships of the municipalities with AEM subsidies to the socio-economic indicators were not proven, it is again appropriate to look at the average values of selected socio-economic indicators and compare them with municipalities that were not awarded any AEM subsidies. These results can be found in Table 4 that suggests that the relative migration balance index reaches a lower value in all selected size categories for municipalities with AEM subsidies in comparison to municipalities without AEM subsidies. In other words, on one hand these municipalities are more stabilized but on the other hand they are less attractive in terms of migration than municipalities without AEM subsidies. In this regard, we can consider the result to be partial proof that AEM subsidies are directed at more problematic municipalities in the Czech Republic. Furthermore, we observe that a higher rate of economically active population is related to a slightly higher unemployment rate in municipalities with AEM subsidies. The employment rate in the primary sector is then logically higher in municipalities with AEM subsidies.

The average values of the index “altitude” are increasing as the size of the municipality decreases, i.e. AEM support provided to municipalities in higher altitudes were awarded to farmers in municipalities with smaller area. In Appendix No. 2, there is a cartogram of AEM subsidies per agricultural area at the level of municipalities (LAU II) which clearly indicates the geographical relationships and even interconnection between AEM subsidies and LFA subsidies. Except for two restricted rural areas (inner peripheries) of municipalities with AEM subsidies, both cartograms in Appendices No. 1 and 2 are similar.

The relationship of AEM measures to relative migration in the given municipalities was low without a significant correlation (−0.027). Fig. 5 illustrates the relationship between altitude and employment rate in the primary sector (0.299). The graph suggests

Table 4
Comparison of selected indicators for municipalities with AEM subsidies and municipalities without AEM subsidies according to selected size categories.

Municipality size category (residents)	Municipalities with AEM subsidies from HRDP						Municipalities without AEM subsidies from HRDP							
	Relative migration balance (%)	EAP (%)	EAP primary sector (%)	Unempl. rate (%)	Average subsidies (CZK)	Average area (ha)	Subsidies/Agr. area	Average altitude	Relative migration balance (%)	EAP (%)	EAP primary sector (%)	Unempl. rate (%)	Average area	Average altitude
Less than 200	1.83	4721	2175	9.65	885,127.69	632.65	2873.53	467.09	1.84	4676	1854	9.54	563.98	416.41
200–500	2.18	4759	1601	9.51	1,725,301.76	1036.31	3279.57	416.01	3.12	4697	1278	8.98	761.48	354.04
500–1000	2.21	4771	1205	9.43	2,592,145.98	1539.80	3270.63	389.92	3.72	4742	797	8.37	969.57	311.30
1000–3000	2.27	4833	877	8.87	3,931,122.02	2,229.73	3604.84	360.73	3.26	4787	536	8.78	1352.48	294.51
More than 3000	0.35	5128	368	8.64	6,767,883.08	3893.22	3793.18	344.45	1.81	4949	246	9.98	1941.10	285.20

Source: Own processing of data from SAIF (2008) and CSO (2008).

that in municipalities that receive AEM subsidies the importance of agriculture in the economy tended to be increasing as the altitude was increasing. Therefore, we assume that AEM subsidies supported alternative forms of agriculture compared to intensive agriculture.

Discussion and conclusion

This paper analyses the relationship between the EU's rural development policy main tools with its territorial cohesion objectives. The analysis of the theoretical and methodological context of the rural development policy led to the conclusion that until the end of the 1980s, rural development had not been a part of research projects. Furthermore, it had not been reflected in relevant policies. The genesis of the rural development policy as an individual strategy was slow and depended upon a number of various factors present within the last two decades. However, the "agricultural" dimension of this policy is still dominating LFA and AEM. These in spite of the postulates of rural economics are complex and not only include agricultural, but non-agricultural activities in rural areas.

In none of the existing analyses, a significant synergy of support of rural development policy and the territorial cohesion objectives has been proved (e.g. Shucksmith et al., 2005; Montresor, 2008; Hrabankova et al., 2006; Viktorova and Pelucha, 2005). We assessed the reasons for these results while we also suggested adjustments of above mentioned analyses. The main added value of this paper is the use of data at the level of municipalities (LAU II). The data on the selected program HRDP CR were analyzed through statistical methods (correlation analysis) with relevant socio-economic data at the level of municipalities.

We analyzed all LFA and AEM payments made during the period 2004–2006 in comparison to the wide range of relevant socio-economic indicators. We used the policy indicators, geographical, demographical and economic indicators. Furthermore, the analysis was focused on a comparison of smaller and bigger municipalities with LFA and AEM payments.

Results central to the analysis identify that a strong correlation between the LFA and AEM support in the Czech Republic and indicators of territorial cohesion are not statistically substantiated. The correlation between LFA and AEM payments and the unemployment rate as well as the relative migration balance appear to be independent. This signifies that the statistical analysis results do not prove that LFA and AEM are aimed at underdeveloped or economically more developed municipalities.

In the case of the AEM, the results of the correlation analyses are similar to the LFA measure outcomes. In terms of the geographical and demographical indicators, the main part of the subsidies is directed at large and to more populated municipalities with areas for agriculture and forestry. In terms of the assessment of the relationship to the unemployment rate, the municipalities that receive LFA or AEM do not indicate a higher rate than municipalities without LFA or AEM subsidies. Therefore, we cannot conclude that in the Czech Republic the LFA and AEM subsidies are directed at richer or less problematic areas. There is only one exception in the case of results related to the rates of an economically active population with employment in the primary sector. The results achieved are, on average, in all municipalities with LFA or AEM subsidies higher than in municipalities without these subsidies. In this sense, we conclude that geographical location and demographical aspects, especially the population size migration stability of the municipality, are more significant than the economic group of indicators

The reason for such results is the delimitation criteria of LFA and AEM in which natural conditions (especially altitude, etc.) were more important than socio-economic. Structural aspects of the rural development policy are not so visible. Considering the period of 2004–2006, the Czech Republic started to spend the EU funding and naturally by little experience from both preparation of programming documents and absorption. The results of the analysis may burden this context, it is then advisable to use the method further at the end of the programming period 2007–2013 for examining the cohesive impact of rural development policy.

The existing conditions for LFA and AEM support are similar to those used during the period 2004–2006. Therefore, we partially conclude that the contemporary “agricultural part” of the rural development policy (CAP second pillar) in the Czech Republic does not represent an important instrument for enhancing the territorial cohesion objectives. The main reason is due to a specific position of the rural development policy in the CAP which emphasizes the agricultural view of rural development.

The structural concept of rural development is still of marginal nature in the CAP and this situation is foreseen for the next programming period 2014–2020 (European Commission, 2010a, p. 5).

Evaluation of the rural development policy measures has recently become important. Therefore we expect higher interest in scientific activities, research projects and policy making in terms of searching for relevant assessment methods. The detailed methodological approach presented in this paper can be further utilized in scientific and research projects aimed at this topic. The necessity of the proposed approach lies in results of the multifunctional agriculture support which are limited in the context of territorial cohesion. The current experience of the Czech Republic presented in this paper demonstrates that positive influence on improvement of territorial cohesion is indirect and ambiguous.

Appendix A.

Fig. A1.

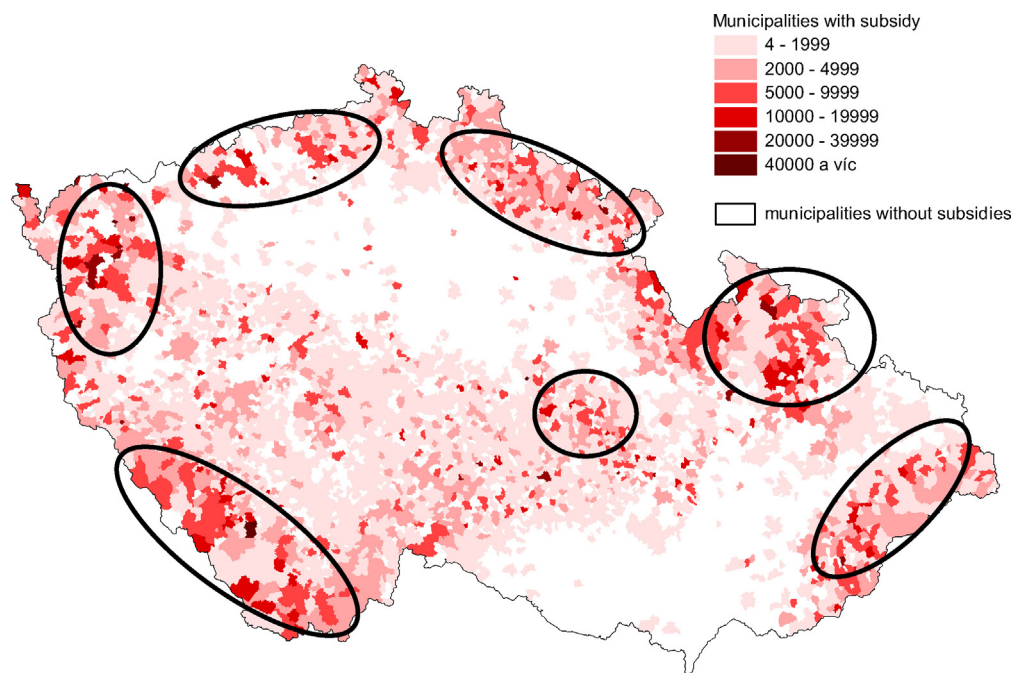


Fig. A1. LFA subsidies per agricultural area at the level of municipalities (LAU II).

Source: Own processing of data from SAIF (2008) and CSO (2008).

Appendix B.

Fig. A2.

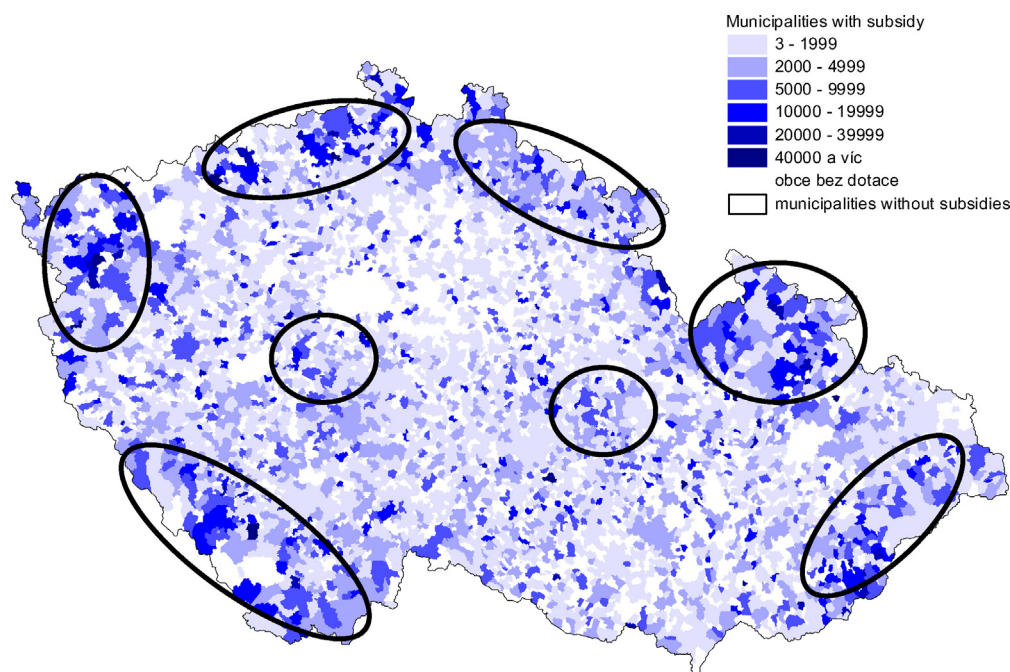


Fig. A2. AEM subsidies per agricultural area at the level of municipalities (LAU II).

Source: Own processing of data from SAIF (2008) and CSO (2008).

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